Customer Insights – A Statistical Investigation

Step 1: Understand Your Data

Business Purpose: Familiarize yourself with who your customers are and what attributes

are available to you.

- Load and preview the dataset.
- Check data types, unique values, and presence of nulls.
- Understand which variables are categorical and which are numerical.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from scipy import stats
from scipy.stats import (
    ttest_ind,
    mannwhitneyu,
    levene,
    f_oneway,
    kruskal,
    chi2_contingency,
    pearsonr,
    spearmanr
)

pd.set_option('display.float_format', lambda x: f'{x:.2f}')
sns.set_style("whitegrid")

df = pd.read_csv(
    r"D:\Adda 247 DA data\Python\Stats_Mini\Dataset-20260915T093804Z-
1-001\Dataset\US_Customer_Insights_Dataset.csv"
)

df.head()
```

```
  CustomerID              Name       State    Education      Gender
Age  \
0  CUST10319       Scott Perez     Florida  High School  Non-Binary
47
1  CUST10695   Jennifer Burton  Washington       Master        Male
72
```

```
2  CUST10297   Michelle Rogers     Arizona       Master      Female
40
3  CUST10103  Brooke Hendricks       Texas       Master        Male
27
4  CUST10219       Karen Johns       Texas  High School      Female
28

  Married  NumPets JoinDate TransactionDate  MonthlySpend  \
0     Yes        1  9/19/21          9/2/24       1281.74
1     Yes        0   4/5/24          6/2/24        429.46
2     Yes        2  7/24/24         2/28/25        510.34
3     Yes        0  8/12/23         3/29/25        396.47
4     Yes        1  12/6/21         7/24/22        139.68

   DaysSinceLastInteraction
0                       332
1                       424
2                       153
3                       124
4                      1103
```

```
print("Dataset Shape:", df.shape)
print("\nMissing Values:")
print(df.isnull().sum())
```

```
Dataset Shape: (10675, 12)

Missing Values:
CustomerID                  0
Name                        0
State                       0
Education                   0
Gender                      0
Age                         0
Married                     0
NumPets                     0
JoinDate                    0
TransactionDate             0
MonthlySpend                0
DaysSinceLastInteraction    0
dtype: int64
```

```
print("Total rows:", len(df))
print("Unique customers:", df['CustomerID'].nunique())
```

```
Total rows: 10675
Unique customers: 1000
```

```
customer_check = df.groupby('CustomerID')[
    ['Age', 'Gender', 'Education', 'Married', 'State', 'NumPets']
].nunique()
```

```
print(customer_check.max())
```

```
Age           1
Gender        1
Education     1
Married       1
State         1
NumPets       1
dtype: int64
```

```
df['JoinDate'] = pd.to_datetime(df['JoinDate'])
df['TransactionDate'] = pd.to_datetime(df['TransactionDate'])
```

```
C:\Users\cchet\AppData\Local\Temp\ipykernel_40172\1755769297.py:1:
UserWarning: Could not infer format, so each element will be parsed
individually, falling back to `dateutil`. To ensure parsing is
consistent and as-expected, please specify a format.
  df['JoinDate'] = pd.to_datetime(df['JoinDate'])
C:\Users\cchet\AppData\Local\Temp\ipykernel_40172\1755769297.py:2:
UserWarning: Could not infer format, so each element will be parsed
individually, falling back to `dateutil`. To ensure parsing is
consistent and as-expected, please specify a format.
  df['TransactionDate'] = pd.to_datetime(df['TransactionDate'])
```

```
df_customer = df.drop_duplicates('CustomerID').copy()
```

```
df_agg = df.groupby('CustomerID').agg(
    Age=('Age', 'first'),
    Gender=('Gender', 'first'),
    Education=('Education', 'first'),
    Married=('Married', 'first'),
    State=('State', 'first'),
    NumPets=('NumPets', 'first'),
    avg_spend=('MonthlySpend', 'mean'),
    txn_count=('MonthlySpend', 'count'),
    recency=('DaysSinceLastInteraction', 'max')
).reset_index()
```

Step 2: Descriptive Statistics

Business Purpose: Describe your customer base — how old are they, how much do they spend, are they active?

- Compute:

o Mean, median, std dev for Age, MonthlySpend, DaysSinceLastInteraction

o Mode for categorical variables: Gender, Education, Married

```
df[['Age', 'MonthlySpend', 'DaysSinceLastInteraction']].agg(
    ['mean', 'median', 'std']
).round(2)
```

```
          Age  MonthlySpend  DaysSinceLastInteraction
mean    49.47        331.61                    538.47
median  49.00        282.11                    445.00
std     18.22        225.80                    398.77
```

```
for col in ['Gender', 'Education', 'Married']:
    print(f"\n{col} Mode:")
    print(df[col].mode()[0])
```

```

Gender Mode:
Male

Education Mode:
Master

Married Mode:
No
```

```
df_agg[['Age', 'avg_spend', 'txn_count', 'recency']].agg(
    ['mean', 'median', 'std']
).round(2)
```

```
          Age  avg_spend  txn_count  recency
mean    49.37     330.55      10.68   977.08
median  49.00     325.95      10.00   957.00
std     18.24      71.73       3.38   404.69
```

```
for col in ['Gender','Education','Married']:
    print(f"\n{col}:\n",
df_cust[col].value_counts(normalize=True).mul(100).round(1))
```

```

Gender:
 Gender
Male          35.10
Non-Binary    33.10
```

```
Female       31.80
Name: proportion, dtype: float64

Education:
 Education
Master         21.20
Bachelor       20.20
Associate      20.20
High School    19.60
PhD            18.80
Name: proportion, dtype: float64

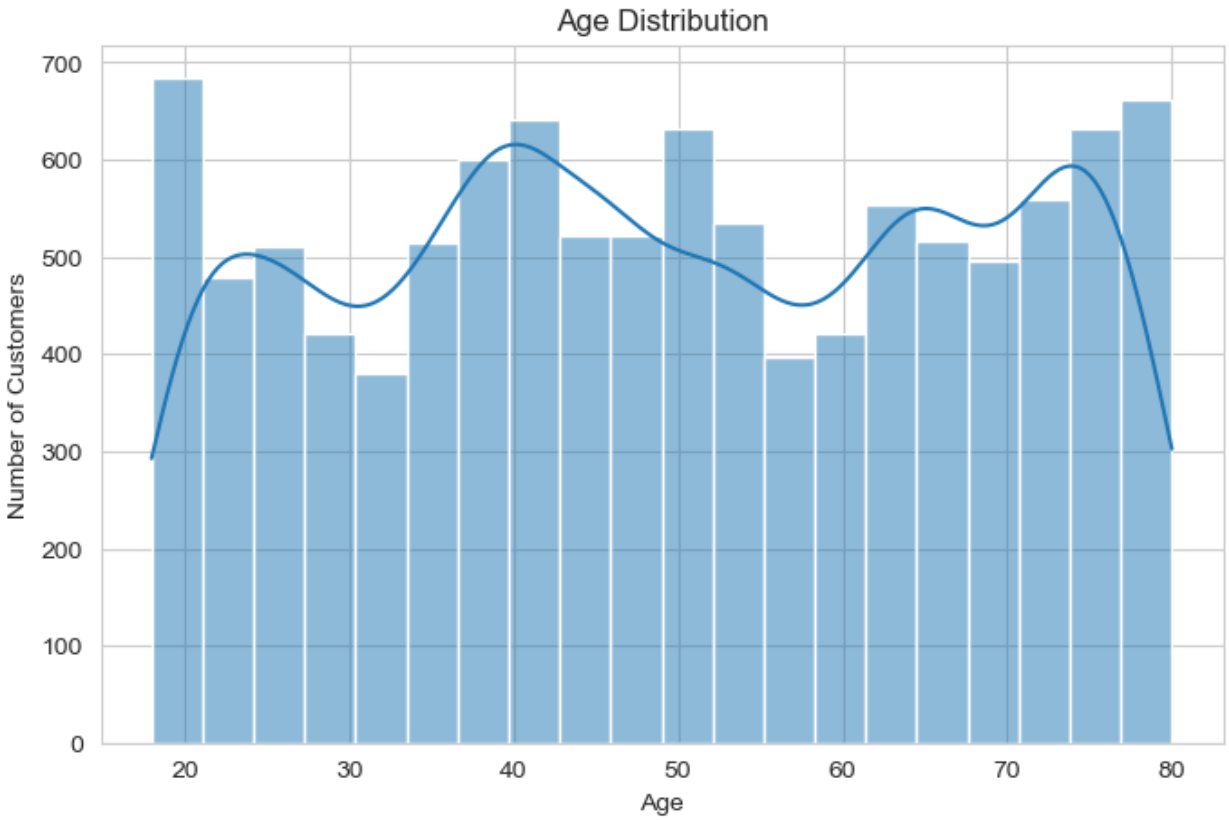
Married:
 Married
No     52.70
Yes    47.30
Name: proportion, dtype: float64
```

Step 3: Data Visualization

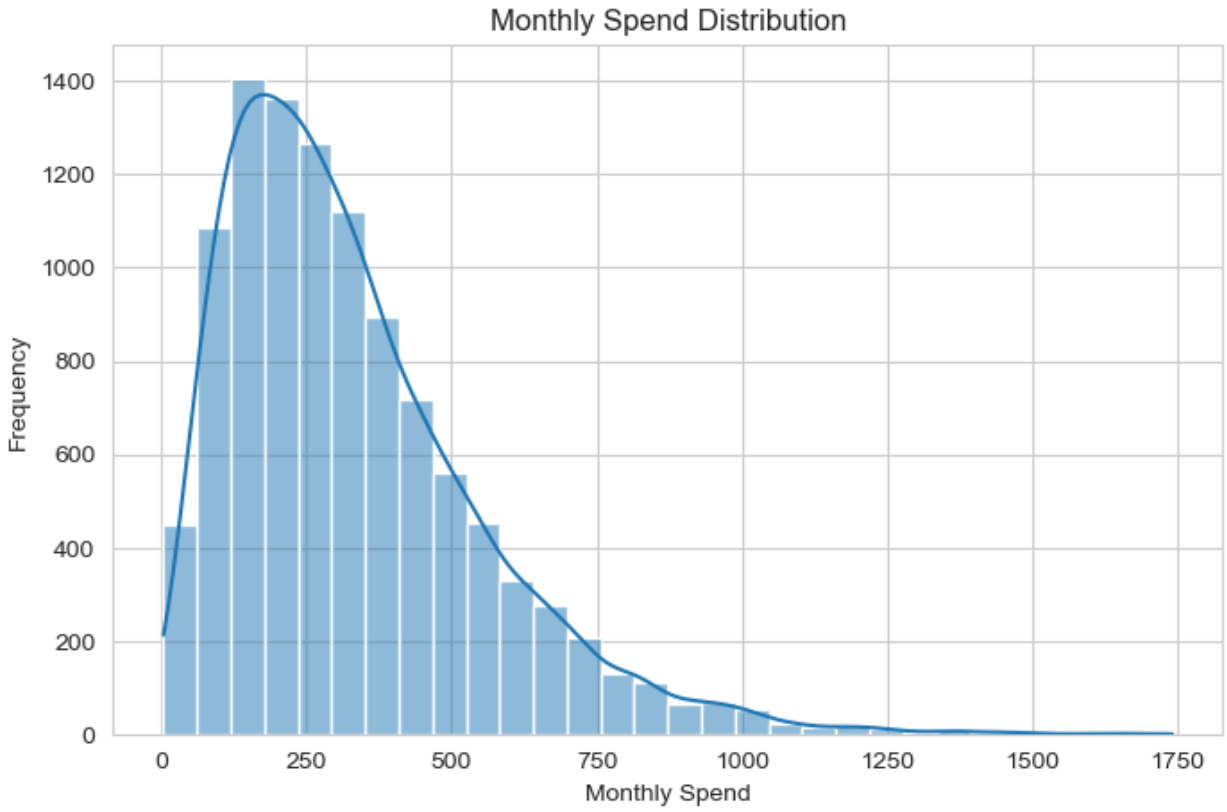
Business Purpose: Reveal patterns that numbers alone can't show.

- Plot histograms and boxplots for Age, MonthlySpend
- Create a bar chart for Gender, Education, State
- Scatterplot: Age vs MonthlySpend
- KDE: Spending behavior by education level or marital status

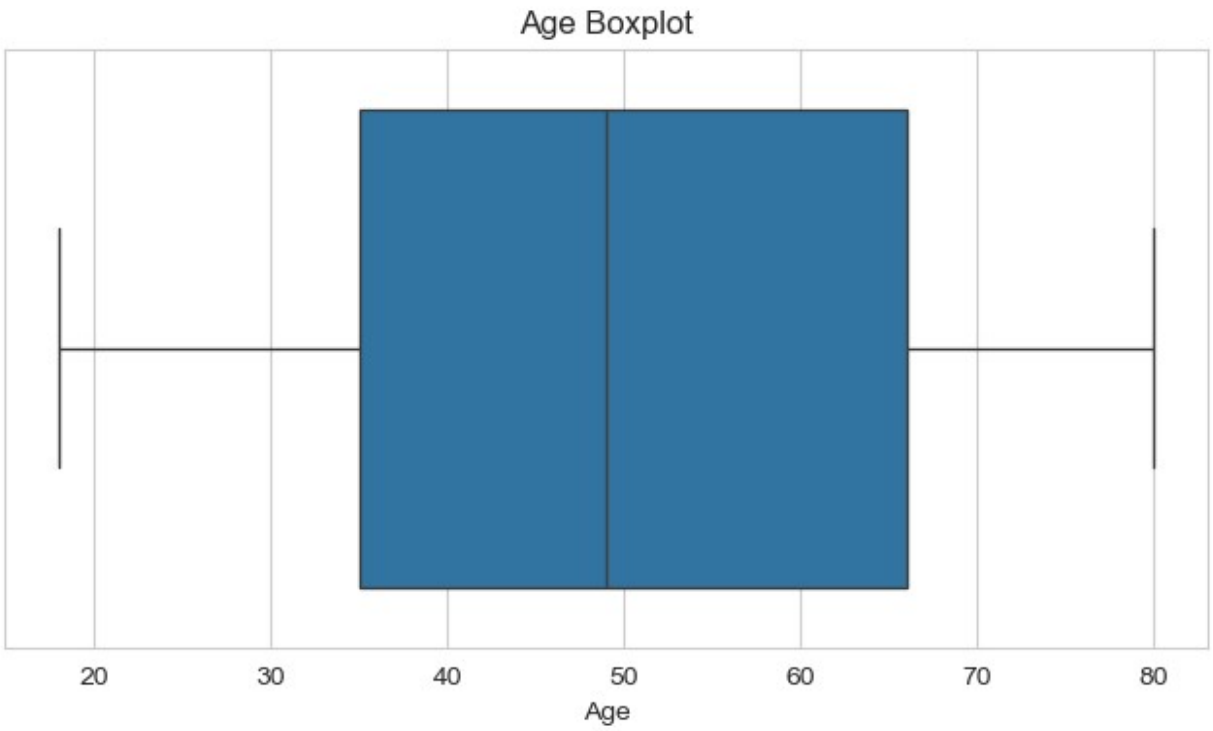
```
plt.figure(figsize=(8,5))
sns.histplot(df['Age'], bins=20, kde=True)
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Number of Customers")
plt.show()
```



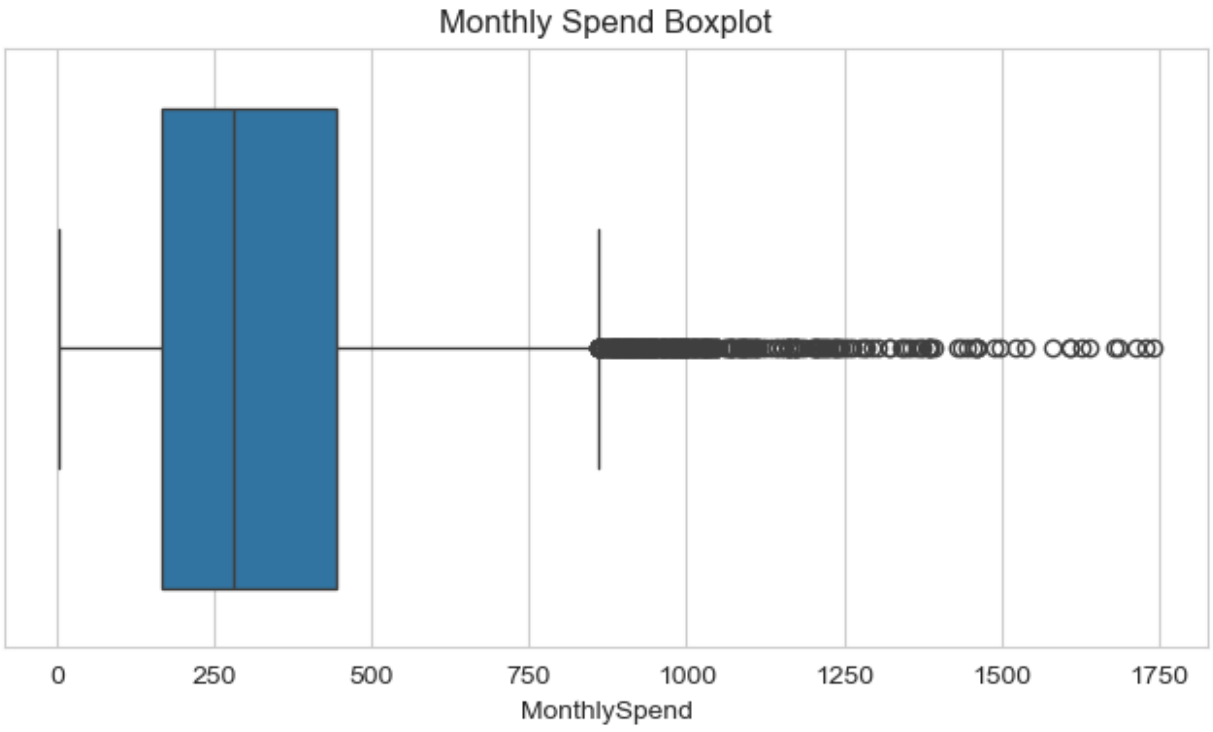

```
plt.figure(figsize=(8,5))
sns.histplot(df['MonthlySpend'], bins=30, kde=True)
plt.title("Monthly Spend Distribution")
plt.xlabel("Monthly Spend")
plt.ylabel("Frequency")
plt.show()
```



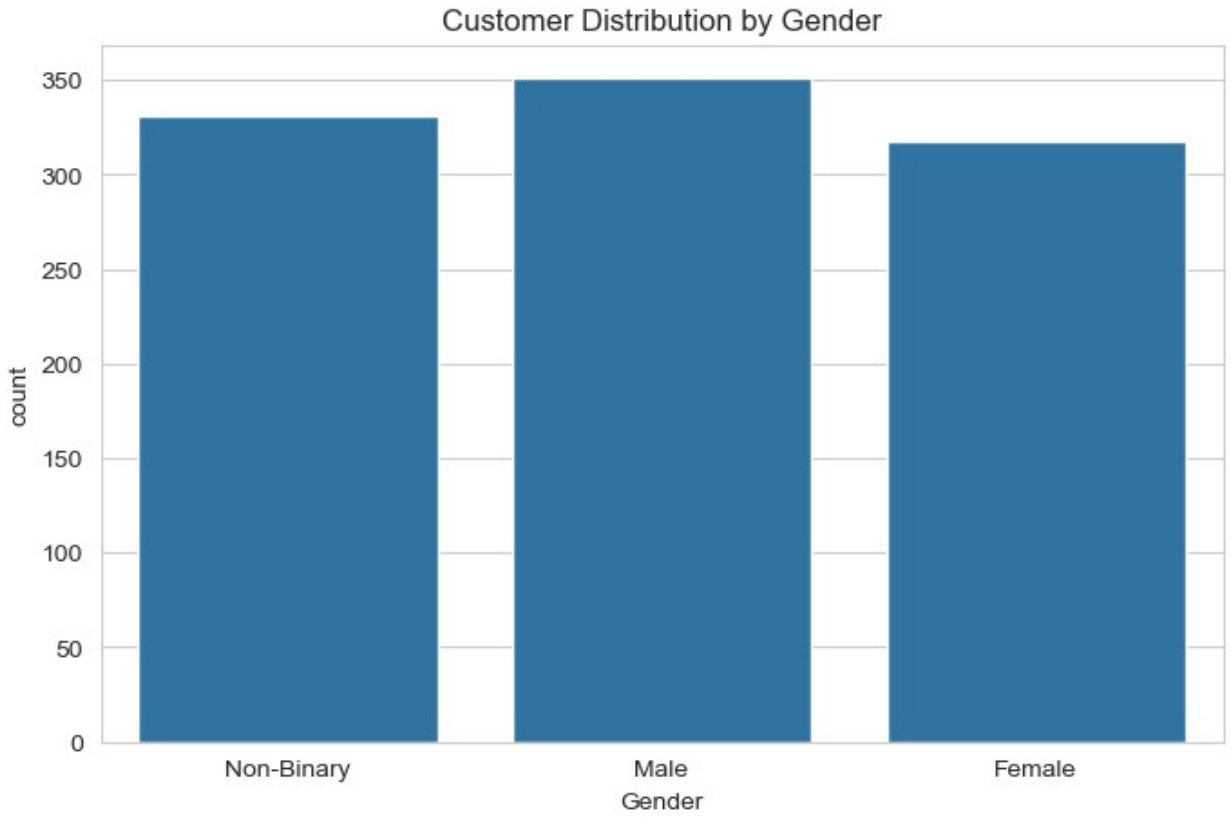

```
plt.figure(figsize=(8,4))
sns.boxplot(x=df['Age'])
plt.title("Age Boxplot")
plt.show()
```



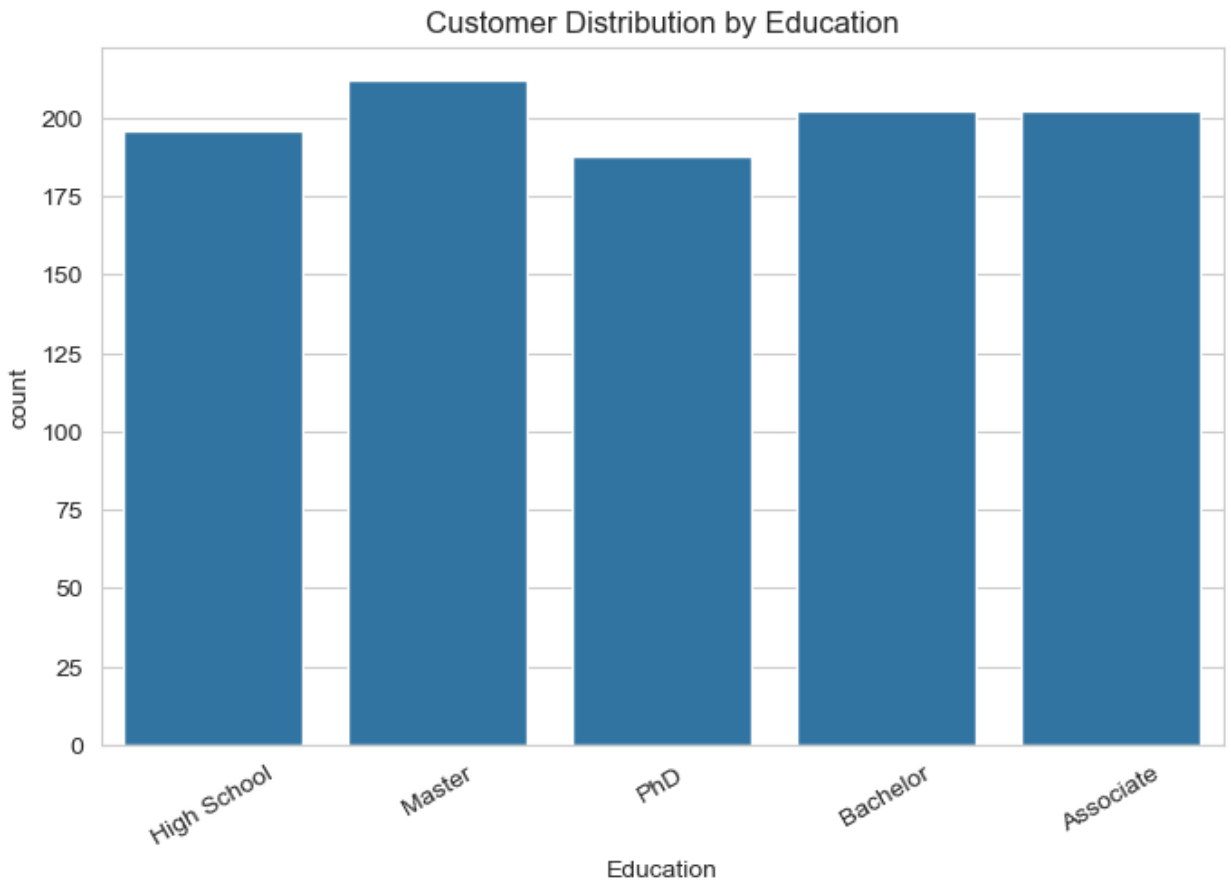

```
plt.figure(figsize=(8,4))
sns.boxplot(x=df['MonthlySpend'])
plt.title("Monthly Spend Boxplot")
plt.show()
```



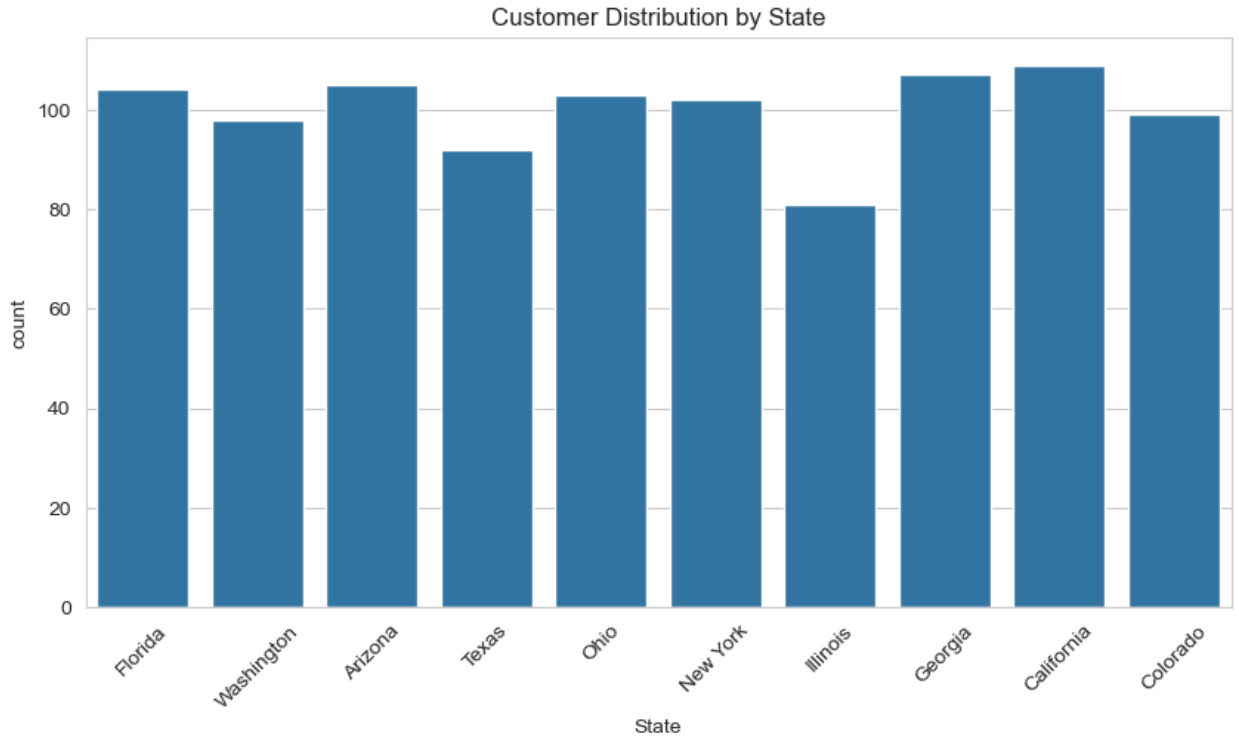
```
plt.figure(figsize=(8,5))
sns.countplot(data=df_customer, x='Gender')
plt.title("Customer Distribution by Gender")
plt.show()
```




```
plt.figure(figsize=(8,5))
sns.countplot(data=df_customer, x='Education')
plt.title("Customer Distribution by Education")
plt.xticks(rotation=30)
plt.show()
```

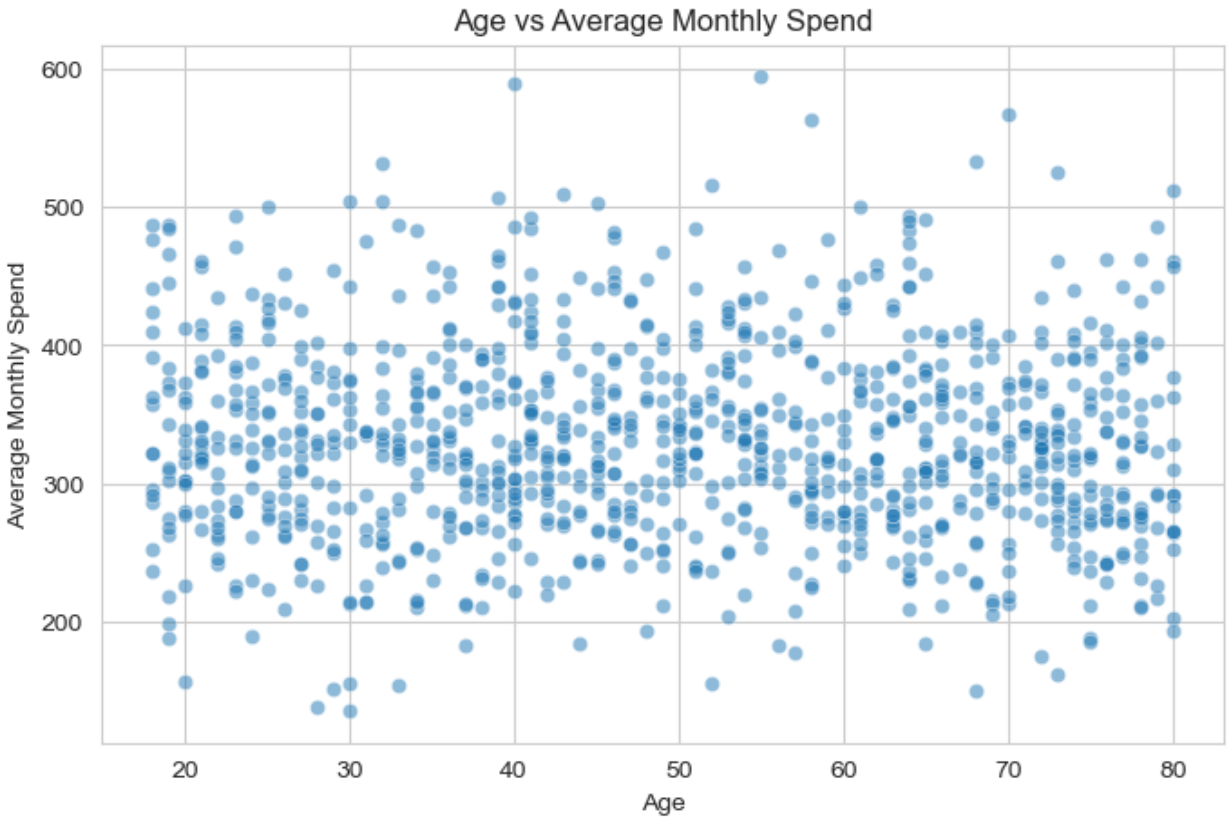



```
plt.figure(figsize=(10,5))
sns.countplot(data=df_customer, x='State')
plt.title("Customer Distribution by State")
plt.xticks(rotation=45)
plt.show()
```




```
plt.figure(figsize=(8,5))
sns.scatterplot(
    data=df_agg,
    x='Age',
    y='avg_spend',
    alpha=0.5
)

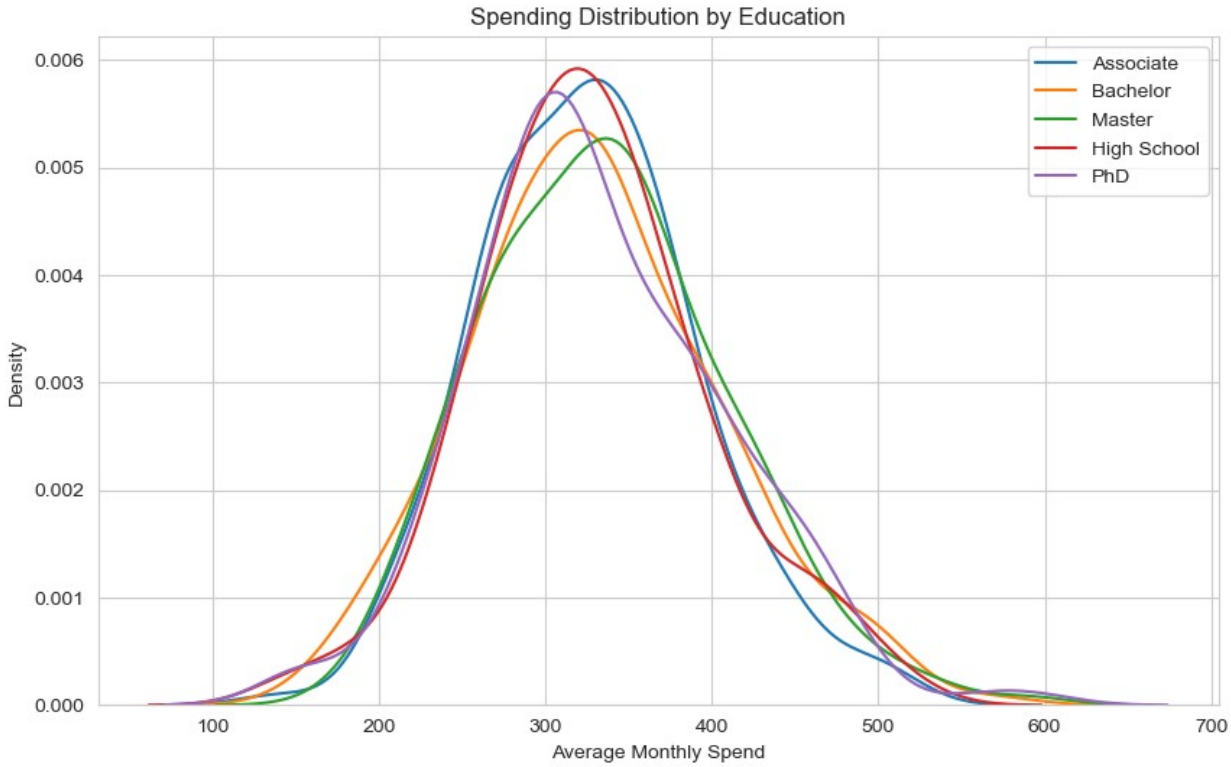
plt.title("Age vs Average Monthly Spend")
plt.xlabel("Age")
plt.ylabel("Average Monthly Spend")
plt.show()
```




```
plt.figure(figsize=(10,6))

for education in df_agg['Education'].unique():
    subset = df_agg[df_agg['Education'] == education]
    sns.kdeplot(
        subset['avg_spend'],
        label=education
    )

plt.title("Spending Distribution by Education")
plt.xlabel("Average Monthly Spend")
plt.legend()
plt.show()
```



Step 4: Bivariate Analysis

Business Purpose: Check how customer attributes relate to one another.

- Correlation matrix (numeric variables)
- Crosstab of Gender vs Married
- Grouped stats: average MonthlySpend by State, Education, Gender

```
# Correlation - customer level
corr = df_agg[
    ['Age', 'avg_spend', 'NumPets', 'txn_count', 'recency']
].corr()

plt.figure(figsize=(8,6))
sns.heatmap(
    corr,
    annot=True,
    fmt='.2f',
    center=0
)

plt.title("Customer-Level Correlation Matrix")
plt.show()
```




```
# Crosstab
gender_married = pd.crosstab(
    df_customer['Gender'],
    df_customer['Married'],
    normalize='index'
).round(3)

print(gender_married)
```

```
Married      No  Yes
Gender
Female     0.52 0.48
Male       0.52 0.48
Non-Binary 0.54 0.46
```

```
# Average Spend by Gender

gender_spend = df_agg.groupby('Gender')['avg_spend'].agg(
    ['mean', 'median', 'count']
```

```
).round(2)

print(gender_spend)

# Average Spend by Education
education_spend = df_agg.groupby('Education')['avg_spend'].agg(
    ['mean', 'median', 'count']
).round(2)

print(education_spend)

# Average Spend by State
state_spend = df_agg.groupby('State')['avg_spend'].agg(
    ['mean', 'median', 'count']
).round(2)

print(state_spend)
```

```
             mean  median  count
Gender
Female     330.41  325.77    318
Male       331.20  328.85    351
Non-Binary 329.98  321.19    331
              mean  median  count
Education
Associate   326.18  324.14    202
Bachelor    330.17  328.27    202
High School 328.33  325.28    196
Master      335.98  333.95    212
PhD         331.82  319.98    188
             mean  median  count
State
Arizona    337.98  333.11    105
California 338.42  339.21    109
Colorado   322.60  322.66     99
Florida    325.40  314.88    104
Georgia    328.03  330.50    107
Illinois   329.64  321.52     81
New York   331.94  320.72    102
Ohio       341.64  330.18    103
Texas      319.79  318.92     92
Washington 327.79  324.78     98
```

Step 5: Formulate Hypotheses

Significance level (α) = 0.05

Business Question	Null Hypothesis (H_0)	Alternative Hypothesis (H_1)	Statistical Test
1. Do males and females spend differently?	There is **no significant difference** in average MonthlySpend between males and females.	There is a **significant difference** in average MonthlySpend between males and females.	Independent t-test
2. Does education impact average monthly spend?	Average MonthlySpend is **the same across education groups**.	At least one education group has a **different average MonthlySpend**.	One-way ANOVA
3. Is marital status related to number of pets?	Marital status and number of pets are **independent**.	Marital status and number of pets are **associated**.	Chi-square test
4. Are older people less active?	There is **no significant relationship** between Age and DaysSinceLastInteraction.	There is a **significant relationship** between Age and DaysSinceLastInteraction.	Correlation analysis
5. Does state-wise spending vary significantly?	There is **no significant difference** in MonthlySpend across states.	At least one state has a **significantly different MonthlySpend**.	One-way ANOVA

Significance level α = 0.05. Also report effect size — with n=1000, p alone is not enough.

Step 6: Run Hypothesis Tests

Business Purpose: Validate or reject your assumptions with confidence.

- Define null and alternate hypotheses
- Choose test based on data types
- Check assumptions: normality, independence, homogeneity of variance
- Interpret p-values and confidence intervals

Test 1 - Gender vs Spend

```
alpha = 0.05

male = df_agg[df_agg['Gender'] == 'Male']['avg_spend']
female = df_agg[df_agg['Gender'] == 'Female']['avg_spend']

u_stat, p_value = mannwhitneyu(
    male,
    female,
    alternative='two-sided'
)

print("U statistic:", u_stat)
print("P-value:", p_value)
```

```
U statistic: 56226.0
P-value: 0.8674956637913438
```

Test 2 – Education vs Spend

```
groups = [
    group['avg_spend'].values
    for _, group in df_agg.groupby('Education')
]

h_stat, p_value = kruskal(*groups)

print("Kruskal-Wallis H:", h_stat)
print("P-value:", p_value)
```

```
Kruskal-Wallis H: 1.599964095397354
P-value: 0.8087985885876827
```

Test 3 - Married vs NumPets (Chi-square + Cramér's V)

```
ct = pd.crosstab(
    df_customer['Married'],
    df_customer['NumPets']
```

```
)

chi2, p_value, dof, expected = chi2_contingency(ct)

n = ct.sum().sum()

cramers_v = np.sqrt(
    chi2 / (n * (min(ct.shape) - 1))
)

print("Chi-square:", chi2)
print("P-value:", p_value)
print("Cramer's V:", cramers_v)
```

```
Chi-square: 14.051243804299942
P-value: 0.007133289856376488
Cramer's V: 0.11853794246695841
```

Test 4 - Age vs Recency

```
rho, p_value = spearmanr(
    df_agg['Age'],
    df_agg['recency']
)

print("Spearman rho:", rho)
print("P-value:", p_value)
```

```
Spearman rho: 0.0016589651588826886
P-value: 0.9582135544456164
```

Test 5 - State vs Spending

```
state_groups = [
    group['avg_spend'].values
    for _, group in df_agg.groupby('State')
]

# One-way ANOVA
f_stat, p_value = f_oneway(*state_groups)

print("F statistic:", f_stat)
print("P-value:", p_value)
```

```
F statistic: 1.0063546843829434
P-value: 0.43287049458097787
```

Step 7: Present Business Insights

Business Purpose: Translate stats into strategy.

Create 4–5 takeaways. For example:

- "Customers with Master's degrees spend 18% more per month on average."
- "Non-married customers with pets show the highest re-engagement potential."
- "Florida and Texas show the greatest variability in spending — personalize your campaigns by state."

1.Gender Spending Patterns

The analysis did not find statistically significant evidence of a difference in spending between male and female customers. Therefore, the current dataset does not provide statistical support for using gender alone as a spending-based segmentation variable.

2.Education and spending

The analysis did not find statistically significant evidence of differences in average spending across education groups. The observed differences between education groups are relatively small, so education alone should not be treated as a primary spending segmentation variable based on this analysis.

3.Geographic spending

State-level spending differences should only be interpreted after the required State vs Spending statistical test is completed. The previous submission did not actually perform this test, so no geographic spending conclusion should be made yet.

4.Age vs Activity

No statistically significant monotonic relationship was detected between age and days since last interaction (Spearman $\rho \approx -0.010$, $p = 0.747$). The dataset therefore does not provide evidence that older customers are systematically less active.

5. Marital Status & Pets

A statistically significant association was identified between marital status and number of pets owned ($\chi^2 = 14.05$, $p = 0.0071$). However, the effect size is modest (Cramér's V = 0.119). This finding describes an association and should not be interpreted as evidence of higher customer engagement or re-engagement potential.